Prabhat Singh

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Skills

Languages & Backend: Python, asyncio, FastAPI, WebSockets, PostgreSQL, SQLAlchemy, Pydantic
Agentic AI & Orchestration: Agent Loop Design, Multi-Agent Systems, Tool/Function Calling, Shared-State Orchestration, MCP, Context Management, Short-Term & Long-Term Memory, Mem0
RAG & Retrieval: RAG Pipelines, Query Routing & Decomposition, Embeddings, Vector Search, Context Grounding, Citation-Based Answers, LLM Output Verification
LLMs & Vector Databases: Gemini, Qwen, GPT-OSS, ChromaDB
Security & Systems: AES-256 Encryption, RBAC, Active Directory/LDAP, Nextcloud API, File System Monitoring
Tools & Platforms: Git, GitHub, Docker, Linux, Claude Code, Postman, pytest

Projects

Multi-Agent RAG System | *Python, asyncio, Gemini, ChromaDB* GitHub

- Built a multi-agent RAG system over PDFs in plain asyncio, no framework: Router, Retriever, Synthesizer and Critic agents coordinate only through a shared state, sequenced by one orchestrator.
- Router classifies queries as no-, single- or multi-retrieval and generates sub-queries; Retriever merges and dedupes ChromaDB results using asymmetric Gemini embeddings and per-page chunking for exact file and page citations.
- Critic verifies each answer's claims against retrieved chunks (Supported / Partially / Unsupported); unsupported answers trigger one bounded retry with a follow-up query, cutting hallucinations without infinite loops.
- Added per-agent tracing and structured JSON outputs with safe fallbacks, so every run shows which agents ran, what was retrieved, and the verdict.

Coding Agent Harness | *Python, OpenRouter, Mem0, MCP* GitHub

- Built Engram, a lightweight terminal-based coding agent that streams LLM responses and executes tools for reading, writing, editing, and running shell commands through a custom agent loop, without external agent frameworks.
- Used OpenRouter as a provider-agnostic model layer, enabling model switching without code changes.
- Integrated GitHub through MCP, allowing the agent to work with existing codebases: browsing repositories, reading and editing files, and managing issues and pull requests.
- Integrated Mem0 for persistent memory, enabling the agent to retain project context and user preferences across sessions.

Experience

Software Developer May 2024 – August 2025
WESEE (Research and Development Lab)

- Built CloudGuard, a FastAPI web application integrated with Nextcloud APIs for secure file uploads, downloads, synchronization and sharing across local and cloud storage.
- Implemented AES-256 file encryption with custom key management and integrated Active Directory/LDAP authentication with role-based access control (RBAC), ensuring only encrypted files were stored in Nextcloud and decryption was restricted to authorized users.
- Developed a watchdog-based folder monitor that automatically encrypts newly added files, with double-click-triggered decryption and timed deletion of decrypted files.

Education

Super30 Program, 100xSchool May 2026 – Present
Selective, cohort-based program focused on building AI/Agentic systems

Scaler Neovarsity Expected Dec 2026
M.Sc. in Computer Science (AI and Machine Learning)

IIT Kanpur 2022
M.Tech in Aerospace (Flight Mechanics and Control)

Galgotias College of Engineering and Technology 2018
B.Tech in Electrical and Electronics Engineering